

PIXEL POLICE DEPARTMENT

Open-world, systems-driven police department sandbox inside a living pixel city · working title · playable prototype, September 2026

PLAY (mobile + desktop): <https://jacobramirezsf.github.io/pixel-police/>
SOURCE: github.com/jacobramirezsf/pixel-police · local:
~/Documents/Makery/projects/PixelPolice

/ 01 / WHAT IT IS

You run a small police department inside a pixel city that does not revolve around you. Roughly 190 simulated residents commute, shop, walk dogs, drink at bars, argue, and occasionally commit crimes — whether or not you're watching. Calls come into dispatch with **incomplete, sometimes wrong information**. You can run everything from above — assigning units, setting policy, managing budget — or drop into any single officer and walk, drive, question, arrest, and shoot in direct control. Every decision feeds a persistent consequence system: neighborhoods remember how they're treated, city hall adjusts your funding, and gangs remember who you arrested.

"There is a little city living its life. I have a police department inside it. Calls are coming in. I can personally respond to something or manage it from above, and over time my decisions determine what kind of department and what kind of city this becomes."

Design principles: emergent stories over scripted missions · dispatch information is imperfect · civilians carry no CRIMINAL/INNOCENT labels · most calls are mundane so violence stays memorable · the player may make bad, reckless, or corrupt decisions — the game answers with consequences, never invisible walls · **mobile-first controls and UI** ("the city is the interface").

/ 02 / THE CITY

- **One generated district** (seeded, deterministic): road grid with crosswalks and working traffic, ~80 buildings with enterable interiors (station, bank, stores, bars, houses, apartments), park, street furniture, day/night cycle with headlights and lamp glow.
- **Four neighborhoods** — wealthy Harbor Heights, commercial Midtown, high-crime Old Docks, residential Cedar Park — each tracking **trust, crime pressure, and tension** that respond to how you police them.
- **Two gang territories:** Dockside 9 and the Eastside Kings each hold a stronghold building and the blocks around it. Territory **hostility** (visible on the GANG TURF map overlay) rises with every member you arrest, wound, or kill — and above ~40, officers walking their turf get jumped. Entering a stronghold uninvited wakes armed defenders.
- **Civilians persist:** homes, jobs, schedules, personalities, criminal records, warrants, concealed weapons, fear, and gang affiliations. They chat, panic, witness, call 911, cooperate or refuse.

/ 03 / DISPATCH & INCIDENTS

Procedural calls, ~1 every 30 seconds of play: noise, traffic stops, suspicious persons, shoplifting, fights, burglaries, robberies, armed robberies, shots fired, vehicle pursuits, gang attacks, bank robberies, shootouts, stronghold raids. What the caller reports and what is actually true **diverge**: weapons go unreported, unarmed people get called in as armed, and calls escalate while units travel ("UPDATE: caller now reports gunshots"). Unanswered calls expire — and cost neighborhood trust and council goodwill. Suspects decide to comply, flee, or fight based on personality, weapon, numbers, and situation. Pursuits have chase-heat (drivers yield or crash under sustained pressure), foot chases end in tackles or turned weapons, and every scene resolution — arrest, warning, citation, gone-on-arrival — is systemic, not scripted.

/ 04 / COMBAT

Free fire, fully unrestricted: draw a weapon and press-and-hold anywhere. Every round is a real ray with spread that stops on walls and trees and hits **whoever is in its path** — suspects, bystanders, other officers. Data-driven weapons (pistols, carbine, shotgun, taser, beanbag; criminal handguns, shotguns, rifles, knives), line-of-sight and cover, injuries from minor to critical to dead, morale-driven surrender and flight, officers who pathfind through doors to threats, EMS and coroner cleanup after scenes quiet down. Gunfire panics crowds and generates its own shots-fired calls — including against you.

/ 05 / RUNNING & GROWING THE DEPARTMENT

Consequences

- Wrongful arrests → complaints, trust loss
- Unjustified shootings → lawsuits (–\$2,000), protests at the station
- Injured officers hospitalized for days; the fallen get End of Watch
- Low morale → resignations; missed calls → crime growth

City hall relationships

- **Council** sets daily funding (0.6× when hostile, 1.4× when supportive)
- **Mayor** unlocks grants and surplus offers
- Both shift daily with resolved calls, arrests, missed calls, lawsuits, deaths, and citywide trust

Earning & upgrading

- **City contracts** (\$900–\$3,800): patrol-hours in a district, crime-reduction pushes, response guarantees, stronghold raids
- **Government surplus**: federal grants, \$800 cruisers, free carbines, ballistic vests for the roster
- **Hire/fire officers** (\$500), buy cars (\$3,000), buy per-officer weapons
- **SWAT team** (\$6,000): 4 heavy officers + van — deploy to any priority call or raid strongholds. Clearing one disbands the gang and drops district crime

The whole loop lives in one thumb-friendly panel: DEPT → OVERVIEW GROW CONTRACTS CITY HALL .

/ 06 / SANDBOX

A first-class cheats menu, not a debug console: money, god mode, infinite ammo, free equipment, heal/spawn officers and cars, free SWAT, gang heat min/max, trust min/max, consequence toggle, time skip — and one-tap

incident spawning up to BANK ROBBERY and LARGE SHOOTOUT. Sandbox use flags saves but never blocks them. Normal play and chaos play are both first-class.

/ 07 / CONTROLS

ACTION	MOBILE (primary platform)	DESKTOP
Camera	drag to pan · pinch to zoom	drag · scroll wheel
Select / orders	tap officer → tap map to move · tap incident to send · MULTI toggle for squads	click · shift-click · same flow
Direct control	CONTROL button → left joystick + action pad (INTERACT / DRAW / RELOAD / CAR / FIRE)	WASD · E interact · F draw · R reload · G car · L lights · Esc release
Shooting	draw, then press-and-hold anywhere to autofire at that point; FIRE button = aim assist	hold mouse button; aim line shows while drawn
Time	pause / 1× / 2× / 4× in top HUD · Space toggles pause · 👁 = clean view (just watch the city)	

/ 08 / THE BUILD

Stack: Vite + TypeScript, plain Canvas 2D, DOM UI, WebAudio-synthesized SFX. **Zero runtime dependencies.** ~119 KB of JS (~40 KB gzipped). Deployed to GitHub Pages by CI (typecheck → build → deploy on every push to main).

MODULE	OWNS
sim/types.ts	Every interface + the single Game state object all systems operate on
sim/data.ts	Data-driven weapon definitions, names, incident titles
sim/world.ts	Seeded city generation (96×72 tiles, neighborhoods, buildings, gang territories, props), A* pathfinding with walk/drive cost functions, Bresenham line-of-sight
sim/agents.ts	Civilian / officer / vehicle spawning + AI state machines, movement, arrests, gang member behavior, SWAT officers, traffic (yield-to-siren, anti-overlap, honking)
sim/incidents.ts	Ambient crime generation with pacing gates, incident lifecycle (queued→assigned→onscene→resolved), imperfect reports, escalation, pursuit chase-heat, gang pressure/ambushes, stronghold raids, EMS sweep
sim/combat.ts	Target-locked AI fire + free-fire point rays (spread, wall clipping, bystander hits), cover, injuries, armor, surrender morale, no-LOS approach via pathfinding
sim/dept.ts	Consequences (casualties→complaints/lawsuits/protests), city relations, contracts, surplus, daily economy, resignations
render/render.ts	Prerendered base-map canvas, dynamic roofs (interiors reveal when entered), entities, tracers, territory/heat overlays, day-night + headlights + lamp glow
ui/ui.ts	Toolbar, sub-tabbed panels, contextual action bar, alerts, control HUD — talks to main via a typed UIApi
main.ts	Game loop, all input (touch/pinch/joystick/keys/hold-to-fire), direct control + car physics, save/load, sandbox implementations
sound.ts	Synthesized shot / radio blip / alarm / chime / thud / honk, distance-attenuated

Architecture invariants

- **The sim never depends on the controlled officer** — direct control is input injection into one entity; everything else keeps simulating.
- All cross-references are **persistent IDs**, never object pointers → saves are plain JSON; the city itself regenerates deterministically from a stored seed (mulberry32).
- Content is data-driven: a new weapon, incident type, contract kind, or unit type is a table entry + small handlers, not a rewrite.
- Dispatch UI never reads an incident's hidden `truth` — only the possibly-wrong `reported` text.

How it was verified

Every pass was playtested headlessly with Playwright driving the real game in Chromium (desktop + emulated iPhone): walking, driving, interacting, arrests, raids, contract payouts, save/load round-trips, long hands-off soaks — asserting on live sim state via a debug hook and screenshotting mobile/desktop/night layouts. Notable bugs this caught: a 36%-of-the-time crash from re-rolling `pick()` inside a `find()` predicate; officers combat-locking on fistfights they couldn't shoot; responders stranded by a 4-pixel vehicle-entry tolerance mismatch; and an "impossible" miss streak that turned out to be bullets legitimately stopping on a tree.

/ 09 / STATE & ROADMAP

Done: living city · dispatch/uncertainty · multi-scale control · driving · pursuits · free-fire combat · injuries/EMS · consequences/protests · gang territory & strongholds · SWAT · contracts/relations/surplus · sandbox · save/load · sound · mobile-first UI · CI deploy.

Next candidates: detectives & investigations (evidence, repeat offenders) · more department policies (pursuit rules, use-of-force) · K9/traffic units · scenario starts (failing department, corrupt department) · multiple maps (small town, county sheriff) · richer interiors · corruption path · audio pass.

Handing off? Read `docs/HANDOFF.md` in the repo first — it carries the architecture invariants, tuning constants, testing recipe, and the known-gaps list, pass by pass.